



TRIOS CYBER

LABORATORY PRACTICE 15

Alert Prioritization and SOC Priority Assessment

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1. Laboratory Objective and Evidence Basis

The activity was designed to prioritize five simulated security alerts using evidence patterns already generated in the authorized laboratory. The alerts were evaluated using four factors required by the exercise: asset criticality, repetition, user impact, and evidence quality. The objective was not to create new attacks, but to convert existing laboratory observations into SOC-style alert scenarios and assign Low, Medium, or High priority based on operational context.

The supporting evidence included repeated failed SSH authentication activity, a failed-to-successful SSH timeline, Apache web access logs, repeated HTTP requests, and a Windows Service Control Manager service-installation event. The priority values below represent simulated analyst prioritization before the activity is confirmed as authorized training traffic.

2. Priority Assessment Criteria

A qualitative prioritization matrix was used so that frequency alone would not determine severity. Low, Medium, and High values were assigned to each factor, and the final alert priority was determined by considering all factors together.

	A	B	C	D
1	Priority Assessment Criteria			
2				
3	Factor	Low	Medium	High
4	Asset Criticality	Disposable/test asset	Standard user or lab endpoint	Critical server or important business endpoint
5	Repetition	Single/isolated event	Several related events	Sustained or rapidly repeated activity
6	User Impact	No observed impact	Potential or limited impact	Confirmed or significant access/impact
7	Evidence Quality	Incomplete or weak evidence	Partially correlated evidence	Clear, timestamped and correlated evidence
8				
9	Priority is assigned by considering all four factors together; a single repeated pattern does not automatically become High priority.			

3. Five Simulated Alert Scenarios

Five alert scenarios were documented in a spreadsheet to provide a consistent comparison of the four required prioritization factors and the resulting SOC priority.

	A	B	C	D	E	F	G	H	I
1	TRIOS CYBER - LABORATORY PRACTICE 15								
2	Alert Prioritization Lab - Five Simulated SOC Alert Scenarios								
3									
4	Alert ID	Simulated Alert Scenario	Evidence Source	Asset Criticality	Repetition	User Impact	Evidence Quality	Priority	SOC Justification
5	ALERT-01	Three failed SSH authentication attempts from the same user and source IP within 29 seconds.	Splunk failed-login correlation evidence	Medium	High	Low	High	Medium	The activity is repeated and strongly supported by correlated evidence, but no successful unauthorized access or direct user impact is present.
6	ALERT-02	A successful SSH login occurs immediately after repeated failed password attempts from the same source.	Linux SSH authentication timeline	Medium	Medium	High	High	High	A success directly following failures can indicate credential guessing followed by account access. The sequence warrants prompt analyst review.
7	ALERT-03	Repeated HTTP 404 responses are generated for /notfound.html on the local Apache test service.	Apache access.log web evidence	Low	High	Low	High	Low	The requests are repeated but target a non-critical local test service and produce no observed impact. The 404 pattern should be reviewed but is low priority.
8	ALERT-04	Repeated GET requests are made to the authorized /test.html resource from the same local client.	Apache access.log request-count evidence	Low	High	Low	High	Low	High repetition alone does not make the activity dangerous. The requests are expected against an authorized test resource and show no adverse user impact.
9	ALERT-05	An unexpected Windows service installation event (Event ID 7045) is detected on an important endpoint.	Windows Event Viewer - Service Control Manager	High	Low	Medium	High	High	Unexpected service creation can represent persistence or unauthorized system modification. Strong host evidence and higher asset criticality justify high priority.

Alert ID	Scenario	Asset	Repetition	Impact	Evidence	Priority
ALERT-01	3 failed SSH attempts from same user/source in 29 s	Medium	High	Low	High	Medium
ALERT-02	Successful SSH login after repeated failures	Medium	Medium	High	High	High
ALERT-03	Repeated /notfound.html requests returning 404	Low	High	Low	High	Low
ALERT-04	Repeated GET requests to authorized /test.html	Low	High	Low	High	Low
ALERT-05	Unexpected Windows service installation - Event 7045	High	Low	Medium	High	High

4. ALERT-01 - Repeated Failed SSH Authentication

The first scenario uses the failed-login correlation evidence in Splunk. Three failed authentication events for user matthew from source 10.10.10.20 occurred inside a 29-second window. Repetition and evidence quality are High because the events were timestamped, grouped by user and source, and validated with the correlation search. The Ubuntu laboratory endpoint is treated as a Medium-criticality asset and there was no successful unauthorized access, so user impact remains Low. The resulting priority is Medium.

The screenshot shows the Splunk Enterprise interface with a new search being created. The search query is as follows:

```
index=main "result=FAILED"
| rex "user={?user}>{?}"
| rex "src={?src}>{?}"
| stats count as failures min(_time) as first_failure max(_time) as last_failure by user src
| eval window_seconds=last_failure-first_failure
| where failures>=3 AND window_seconds<=60
| convert ctime(first_failure) ctime(last_failure)
| table user src failures first_failure last_failure window_seconds
```

The search results show 6 events. The table below displays the results for user matthew from source 10.10.10.20.

user	src	failures	first_failure	last_failure	window_seconds
matthew	10.10.10.20	3	09/12/2026 13:00:01	09/12/2026 13:00:30	29

5. ALERT-02 - Successful SSH Login after Repeated Failures

The second scenario uses the SSH authentication timeline where two failed password attempts were followed by an Accepted password event from the same Windows source address. In an operational SOC, this sequence can indicate credential guessing followed by successful account access. Evidence quality is High because the same account, source IP, service, client port, and timestamps are present in the journal. Because successful access can directly affect the user account, user impact is assessed as High and the final priority is High.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. Todos los derechos reservados.

Prueba la nueva tecnología PowerShell multiplataforma https://aka.ms/pscore6

PS C:\Users\matthew> ssh matthew@10.10.10.10
matthew@10.10.10.10's password:
Permission denied, please try again.
matthew@10.10.10.10's password:
Permission denied, please try again.
matthew@10.10.10.10's password:
Welcome to Ubuntu Focal Fossa (development branch) (GNU/Linux 5.4.0-216-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

770 updates can be installed immediately.
488 of these updates are security updates.
Para ver estas actualizaciones adicionales, ejecute: apt list --upgradable

Last login: Tue Sep  8 22:51:49 2026 from 10.10.10.20
matthew@matthew-VirtualBox: $
```

```
matthew@matthew-VirtualBox:~$ sudo journalctl -u ssh --since "10 minutes ago" -o short-iso
| grep -Ei "Accepted password|Failed password"
2026-09-12T12:39:38-0500 matthew-VirtualBox sshd[3876]: Failed password for matthew from 1
0.10.10.20 port 49673 ssh2
2026-09-12T12:39:45-0500 matthew-VirtualBox sshd[3876]: Failed password for matthew from 1
0.10.10.20 port 49673 ssh2
2026-09-12T12:39:49-0500 matthew-VirtualBox sshd[3876]: Accepted password for matthew from
10.10.10.20 port 49673 ssh2
```


6. ALERT-03 - Repeated HTTP 404 Responses

The third scenario is based on Apache access-log evidence. The local browser repeatedly requested /notfound.html and Apache returned HTTP 404 responses. Repetition and evidence quality are High because the access log preserves the client IP, method, path, status code, and user-agent. However, the service is a non-critical local test service and no user or system impact was observed. The alert is therefore assigned Low priority despite the repeated pattern.

```
matthew@matthew-VirtualBox:~$ sudo grep '"GET ' /var/log/apache2/access.log
127.0.0.1 - - [15/Sep/2026:18:51:09 -0500] "GET /test.html HTTP/1.1" 200 311 "-" "Mozilla/5.0 (X
11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:51:09 -0500] "GET /favicon.ico HTTP/1.1" 404 487 "-" "Mozilla/5.0
(X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:51:14 -0500] "GET / HTTP/1.1" 200 3477 "-" "Mozilla/5.0 (X11; Ubun
tu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:51:14 -0500] "GET /icons/ubuntu-logo.png HTTP/1.1" 200 3623 "http:
//127.0.0.1/" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:52:59 -0500] "GET /notfound.html HTTP/1.1" 404 488 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:20 -0500] "GET /notfound.html HTTP/1.1" 404 488 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:20 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:21 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:21 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:22 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:23 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:25 -0500] "GET / HTTP/1.1" 200 3476 "-" "Mozilla/5.0 (X11; Ubun
tu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:25 -0500] "GET /icons/ubuntu-logo.png HTTP/1.1" 304 180 "http:/
/127.0.0.1/" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:26 -0500] "GET /test.html HTTP/1.1" 304 179 "-" "Mozilla/5.0 (X
11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:27 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:28 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
```

```
matthew@matthew-VirtualBox:~$ sudo grep ' 404 ' /var/log/apache2/access.log
[sudo] contraseña para matthew:
127.0.0.1 - - [15/Sep/2026:18:51:09 -0500] "GET /favicon.ico HTTP/1.1" 404 487 "-" "Mozilla/5.0
(X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:52:59 -0500] "GET /notfound.html HTTP/1.1" 404 488 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:20 -0500] "GET /notfound.html HTTP/1.1" 404 488 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:20 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:21 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:21 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:22 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:23 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:27 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:28 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:55:42 -0500] "GET /favicon.ico HTTP/1.1" 404 487 "-" "Mozilla/5.0
(X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
```


7. ALERT-04 - Repeated Requests to an Authorized Test Resource

The fourth scenario uses the request-count evidence from the same Apache laboratory. Repeated GET requests were observed for the authorized /test.html resource. This scenario demonstrates why repetition alone must not automatically produce a High alert: the destination is an expected test page, the asset criticality is Low, no user impact is present, and the evidence clearly explains the activity. The final priority is Low.

A focused grep for /test.html confirmed two entries: an initial 200 response and a later 304 response

```
matthew@matthew-VirtualBox:~$ sudo awk '{print $7}' /var/log/apache2/access.log | sort | uniq -c
| sort -nr
  9 /notfound.html
  3 /icons/ubuntu-logo.png
  3 /
  2 /test.html
  2 /favicon.ico
```

```
matthew@matthew-VirtualBox:~$ sudo grep '/test.html' /var/log/apache2/access.log
127.0.0.1 - - [15/Sep/2026:18:51:09 -0500] "GET /test.html HTTP/1.1" 200 311 "-" "Mozilla/5.0 (X
11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:26 -0500] "GET /test.html HTTP/1.1" 304 179 "-" "Mozilla/5.0 (X
11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
```

8. ALERT-05 - Windows Service Installation Event 7045

The fifth scenario uses the Windows System log, where Service Control Manager Event ID 7045 recorded a service installation. For prioritization, the scenario assumes the event occurs unexpectedly on an important Windows endpoint. An unexpected service can indicate persistence or unauthorized system modification, and Event Viewer provides strong host-based evidence. Even without high repetition, the combination of higher asset criticality, potential user or system impact, and High evidence quality results in a High priority.

visor de eventos

Archivo Acción Ver Ayuda

Visor de eventos (local)

- Vistas personalizadas
- Registros de Windows
 - Aplicación
 - Seguridad
 - Instalación
 - Sistema
- Eventos reenviados
- Registros de aplicaciones y s
- Suscripciones

Sistema Número de eventos: 472

Filtros: Registro: System; Origen: ; Id. del evento: 7036,7045. Número de eventos: 3

Nivel	Fecha y hora	Origen	Id. del evento	Categoría de l...
Información	3/9/2026 23:28:14	Service Contr...	7045	Ninguno
Información	3/9/2026 23:26:03	Service Contr...	7045	Ninguno
Información	3/9/2026 23:23:23	Service Contr...	7045	Ninguno

Acciones

Sistema

- Abrir registro guardad...
- Crear vista personaliza...
- Importar vista personal...
- Vaciar registro...
- Filtrar registro actual...
- Borrar filtro
- Propiedades
- Buscar...
- Guardar archivo de reg...
- Adjuntar tarea a este re...
- Guardar filtro en vista ...
- Ver
- Actualizar
- Ayuda

Evento 7045, Service Control Manager

General Detalles

Se instaló un servicio en el sistema.

Nombre de registro: Sistema

Origen: Service Control Manager Registrado: 3/9/2026 23:28:14

Id. del: 7045 Categoría de tarea: Ninguno

Nivel: Información Palabras clave: Clásico

Usuario: SYSTEM Equipo: WIN-HF1UE3GRE4

Crea un filtro.

Escribe aquí para buscar.

Red no identificada Sin acceso a Internet 26

Evento 7045, Service Control Manager

General Detalles

Nombre de registro: Sistema

Origen: Service Control Manager Registrado: 3/9/2026 23:28:14

Id. del: 7045 Categoría de tarea: Ninguno

Nivel: Información Palabras clave: Clásico

Usuario: SYSTEM Equipo: WIN-HF1UE3GRE4

Código de operación: Información

Más información: [Ayuda Registro de eventos](#)

9. Alert Prioritization Summary

The comparison shows that priority depends on context rather than on event count alone. ALERT-03 and ALERT-04 contain repeated activity but remain Low because they affect a local test service and have no meaningful user impact. ALERT-01 is Medium because the repeated authentication failures deserve investigation but did not result in access. ALERT-02 and ALERT-05 are High because successful account access after failures and unexpected service installation can create substantially greater security impact.

Alert	Key Evidence	Priority	Reason
ALERT-01	3 failures / same user + source / 29 s	Medium	Repeated authentication pattern with strong evidence but no successful access.
ALERT-02	Failed, failed, then Accepted password	High	Potential credential compromise because authentication succeeds after failures.
ALERT-03	Repeated 404 for /notfound.html	Low	Repeated but confined to a low-criticality test service with no impact.
ALERT-04	Repeated GET to /test.html	Low	Expected authorized resource; repetition has no adverse effect.
ALERT-05	Windows Event ID 7045 service installation	High	Unexpected persistence-capable change on an important endpoint.

10. Conclusions

The alert-prioritization exercise was completed using five simulated SOC scenarios derived from laboratory evidence. Each alert was reviewed against asset criticality, repetition, user impact, and evidence quality before a Low, Medium, or High priority was assigned. The exercise demonstrated that repeated activity is only one part of prioritization: context, impact, asset importance, and the reliability of the available evidence determine which alerts should receive immediate analyst attention. The resulting spreadsheet provides a concise triage reference that can be reused for future sample alerts.